

JACOB FAIN

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EDUCATION

Harvey Mudd College, Claremont, CA • GPA 3.62

Expected May 2026

B.S. Engineering

Relevant Coursework

Senior Capstone • Mechanical Design • Manufacturing Planning and Execution • Dynamics of Rigid Bodies • Continuum Mechanics • Thermodynamics • Materials Engr. • Applied Math for Engr. • Advanced Systems Engr. and Controls • Electronic and Magnetic Circuits / Devices • Engr. Management • Statistics

Skills

Engineering Software: SolidWorks, MATLAB / Simulink, Excel, MotionGenesis Kane, AutoCAD, LabVIEW, COMSOL
Machine Shop / Laboratory: Lathe, Mill, Waterjet, Laser Cutter, MIG Welding, Power Tools, Oscilloscope, Soldering
Programming Languages: Python, Arduino, HTML / Markdown

ENGINEERING EXPERIENCE

Machine Shop Proctor, Harvey Mudd College

August 2024 - Present

- Training fellow students on safe and sanitary operation of various metal and wood fabrication machines. Fostering a welcoming, safe learning environment fundamental to countless student projects and experiences.
- Advising DFM and fabrication of dozens of parts, fixtures, and assemblies for classes, clubs, research teams, etc.

Student Mechanical Engineer, Department of Water and Power, Los Angeles, CA

June 2025 - August 2025

- Drafted scope-of-work package detailing project background, design and procurement status, and labor required to install a turbine bypass line at Division Creek Power Plant. Bypass will facilitate constant flow and safer maintenance.
- Created a system curve and static pressure calculator to determine bypass line valve and piping specifications. Visited Division Creek to redline P&ID and site plan drawings for scope-of-work package. Corresponded with valve vendors.
- Inspected 3 other facilities, enabling associates to update P&ID, HVAC, and site plan drawings across 4 projects.

ENGINEERING PROJECTS

Clinic Engineer, Harvey Mudd-Medtronic Clinic

January 2025 - May 2025

- Worked with 5 student engineers under Medtronic liaisons and a faculty advisor to design, fabricate, and deliver a precision O-ring lubrication solution for Medtronic's Guardian Sensor 3 continuous glucose monitoring device.
- Created prototypes in SolidWorks for safety, ergonomics, and manufacturability. Redesigned a parameterized jig from scratch, enabling 5+ rapid iterations. Prepared technical drawings and machined 304 stainless steel to 0.001" tolerances.

Vice President & Mechanical Team Lead, Mudd RoboSub Club

October 2022 - Present

- Developing subsystems across 2 modular autonomous underwater vehicles for the international RoboSub competition.
- Updating vehicle design via CAD and FEA for robustness, machinability, and buoyancy. Computing thruster moments for maneuverability and O-ring / pre-load specifications for watertightness via Excel. Reconfiguring outdated files.
- CADed launcher to separately fire 2 torpedoes, using a single servo holding springs in compression via prongs.

Structures Team Lead, Mudd Amateur Rocketry Club

September 2022 - Present

- Guiding 10+ peers through rocket CAD, fabrication, and assembly. Writing design and fabrication best practices, such as process routers for machining composite structural components to 0.001" tolerances via mill, waterjet, and router.
- Assembled, launched, and successfully recovered a rocket at the 2025 FAR-Unlimited competition, placing 3rd.
- Developed novel bulkheads and aft airframe assembly, greatly increasing internal load transfer. Verified design via FEA.
- Created parachute and mounting system for a remote-control rover intended to deploy on descent.